



LEELA EDUCATION SOCIETY

(By the Teachers, By the Teachers, For the Students)

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Everything we do, We do it for you



G.V ACHARYA INSTITUTE OF ENGINEERING & TECHNOLOGY

Veeracharya Technical Education Campus, Opp. Shelu Railway Station, (E), 2nd Station After Badlapur Pin - 410 101

Student Name: Manan Vernekar Subject Python Date _____

Start From here

Complete Guide

- Requirements -

Python + (VS code (optional)) as editor.

- * Basics

- print("") - to print

- Python is case-sensitive

- variables = values

- type() = returns variable type.

this are all
'functions'

- str = string · int = integer · float = decimal values.

- boolean (bool) = true/false.

21:00

- input("") - to take input from user.

↘ by default collects info in 'string'

- print("Hello" + name)

↑ concatenation (combine two strings / variables)

- # to comment.

27:00

- use int() or float() to convert specific values to it.

↘
Type casting → done by programmer. (explicit)

- Type conversion → done by interpreter. (implicit)

* Strings

- strings can be stored in ' ' or " "

• upper() - to uppercase the string.

• lower() - to lowercase.

- Strings are immutable (cannot change later) using the above operations original string don't change, it returns new string.

- Can create new string with changed information, but cannot change the original.

- use cmd + / to comment selected content.

- find() - used to find the index (position) of the string.
index starts from 0, 1, 2, 3...

String
operations

- replace() - to replace string ^{mentioned} content to another mentioned content.

- print('P' in name)
to check presence

- There are some reserved keywords like True, in, False, for, if etc in Python.

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* Arithmetic operators - for mathematical operations.

+ , - , / , *
↓

• Use '//' for integer value for decimal answer.

• % - remainder values returned
modulo

• Use '**' for power operator i.e. $5^{**}3 \rightarrow 5^3$

* Assignment operators -

• += to add some value to existing value

-= , *= , /= are some assignment operators.

* Operator Precedence.

ex -

• *, / have greater priority than +, -.

• (,) have the highest.

* Comparison Operators

> , < , >= , <= ... used to compare and return True/False values.

!= - not equal to == equal to

* Logical Operators - OR, AND, NOT.

returns the final Boolean value based on 'or', 'and', 'not' operators conditions.

* Conditional Statements

- give 'if' conditions to return any further results based on if the condition is satisfied.

* indentation - used in python for proper spacing in code lines.

- if In python after 'if' condition ':' is used to consider all the lines of code after that in indentation of the statement to be executed thereafter.

- 'if, elif' can be used for conditions, like...

↓
if this condition or then this condition.

- 'if, elif, else:' can be used like first 'if' condition, then 'elif' conditions or 'else' condition.

* Range

- $\text{range}(n) = 0 \text{ to } n-1$, are returned in this range.
- ex- $\text{range}(1, 8) \rightarrow$ will return range from 1 to 7 ($n-1$ or $8-1$).
- by default $\text{range}(\text{start}=0, \text{stop}, \text{step}=1)$.



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* Loops-

• while loop is used for creating loop that executes same statements if and until condition satisfies.

• For loop creates loop until and if the condition satisfies and especially in range.

• ex - for i in nums → creates loop in 'nums' list/sequence. it will print all 'i' iterations in the 'nums' range

* Break - To stop the loop at a condition if it gets met.

* Continue - It is used ~~to~~ in a range to skip the condition that is met and continue with the loop ahead.

* Data Types ↓

* Primitive - int, bool, float, str.

* Non-Primitive -

* List - can make a list of things.

↳ uses - can be of different type of items.
- is mutable (can be edited)

• to slice list

→ list [start:end] → not included
↳ start.

• append - used to add items/values in last/end of list.

• insert - to insert value at specific index.

• clear - to empty the list.

* Tuple - are immutable.

tuple() _{uses} - can be used to create tuple, but also without ().

mostly similar functions like List. ^(Operations)

• .count() - can be used for counting number of times items are there in.

* Set - collection of unique items.
uses { }

• set don't store duplicate items in it.

• use .len() - for length of set without including duplicate items.

* Dictionary - collection of key-value pairs.
uses {key: value}.

• Dictionary is mutable.

• Items can be added or modified.

• key and values can be accessed and printed separately.

* We can create data types like tuple inside another data type like list.

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* Function is used to simplify tasks, that we have to repeat many times here / there in code.

eg- def function-name^(*): to define.
 ← parameters.

• function-name^(x) to call.
 ← arguments

* User-defined functions are defined by user.

* In-built functions like - print, len, min, max, etc.

* Module functions is bundle of functions like math.

* There are various modules in Python.

* from module_name import functionname1, functionname2, etc
 to import module functions.

* functions can also be called like math.sqrt() math.log2()

* From random module ↴

• random.random() - values randomly generated
b^twn >0 and <1

• random.randint(a,b) - to generate values in range.

* return is used to exit function and pass a result back to the program that called it.